13/oct/2026

- Databases stores data so that they, or another application, can find it again later.
- Caches remembers the results of an expensive operation, to speed up reads.
- Search indexes allows users to search data by keyword or filter it in various ways
- Stream processing Send a message to another process, to be handled asynchronously
- Batch processing. periodically crunch a large amount of accumulated data

Datastores that are used as message queues — Redis

NOTES There are message queues with database like durability guarantees — Apache Kafka

- Memcached helps in application-Managed caching layer
- Elasticsearch or Solr helps in full text search Server

Questions while designing a data system or Service?

1) How do you ensure that data remains correct and complete, even when things go wrong internally?

2) How do you provide consistently good performance to clients, even when part of your system are degraded?

3) How do you scale to handle an increase in load?

4) What does a good API for the Service look like?

NOTES

21 FEBRUARY
THURSDAY

FEBRUARY '13

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28			

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- Three most important concerns for software systems

1) Reliability
2) Scalability
3) Maintainability

Reliability

For software it means
1) The application performs the function that the user expected

2) It can tolerate the user making mistakes or using the software in unexpected ways

3) Its performance is good enough for the required use case, under the expected load and data volume

4) The system prevents any unauthorised access and abuse

NOTES

One line — It should work correctly, even when things go wrong



- Things that can go wrong are called faults, and Systems that anticipate faults and can Cope with them are called fault-tolerant or resilient.

- fault is not same as failure — Remember this
- A fault is usually defined as one component of the System deviating from its spec, whereas a failure is when the System as a whole stops providing the required service to the user.
- Read about → The Netflin Chaos Monkey

Faults that Can be cured

i) Hardware faults

- Hard disk Crashes
- the power grid has a Blackout
- RAM becomes faulty
- Someone unplugs the wrong network Cable

* MTTF - Mean time to failure

2) Software Errors

- A software bug that causes every instance of an application server to crash when given a particular bad input

e.g. Consider the leap second on June 30, 2012, that caused many applications to hang simultaneously due to a bug in linux kernel.

- A runaway process that uses up some shared resource - CPU time, memory, disk space, or network bandwidth.

- A service that the system depends on that slows down, become unresponsive or starts returning corrupted responses

- Cascading failures, where a small fault in one component triggers a fault in another component which in turn triggers further faults.

NOTES